

`-set-file` accepts input from a file or *stdin* and the `-` (dash) specifies the use of *stdin*. In this case, File B will have the same ACL settings as File A.

Setting an explicit ACL mask

An ACL mask can be explicitly set on a file or directory to limit the maximum effective permissions for named users, the group owner, and named groups. This restricts any existing permissions that exceed the mask, but does nothing to permissions that are less permissive than the mask.

```
# setfacl -m m: :r file
```

This adds a mask value that restricts any named users, the group owner and any named groups to read-only permission, regardless of their existing settings. The file owner and other users are not impacted by the mask setting. *getfacl* will show an ‘effective’ comment beside entries that are being restricted by a mask setting.

 **Note:** By default, the ACL mask is recalculated each time one of the impacted ACL settings (named users, group owner, or named groups) is modified or deleted, potentially resetting a previous explicit mask setting. To avoid mask recalculations, use `-n` or include a mask setting (`-m m::perms`) with any *setfacl* operation that modifies mask-affected ACL settings.

Recursive ACL modifications

When setting an ACL on a directory, it is common to want to apply the ACL recursively to the directory structure and files. Use the `-R` option to do this. The `X` (upper case X) permission is often used with recursion, so that files with the *execute* permissions set retain the setting, and directories get the *execute* permissions set to allow directory search. It is considered good practice to also use the uppercase `X` when non-recursively setting ACLs, as it prevents an administrator from accidentally adding *execute* permissions to a regular file.

```
# setfacl -x u:name, g:name file
```

This only removes the named user and the named group from the list of file or directory ACLs. Any other existing ACLs remain active.

It is possible to use the delete (`-x`) and modify (`-m`) operations in the same *setfacl* operation.

The mask can only be deleted if there are no other ACLs set (excluding the base ACLs which cannot be deleted), so it must be deleted last. The file will no longer have ACLs and *ls-l* will not show the ‘+’ symbol next to the permissions string. Alternatively, to delete all ACLs on a file or directory (including default ACLs on directories), use the following command:

```
# setfacl -b file
```

Controlling default ACL file permissions

A directory can have default ACLs set on it that are automatically inherited by all new files and new sub-directories. There can be default ACL permissions set for each of the standard ACL settings, including a default mask.

A directory still requires standard ACLs for access control because default ACLs do not implement access control for the directory—they only provide ACL permission inheritance support.

Here is an example:

```
# setfacl -m d:u:name:rx directory
```

This adds a default named user (*d:u:name*) with read-only permission and *execute* permission on sub-directories.

The *setfacl* command for adding a default ACL for each of the ACL types is exactly the same as for standard ACLs, but prefaced with *d:*. Alternatively, use the `-d` option on the command line.

 **Important:** When setting default ACLs on a directory, ensure that users will be able to access the contents of new sub-directories created in it by including the *execute* permissions on the default ACL. Users will not automatically get the *execute* permissions set on newly created regular files because, unlike new directories, the ACL mask of a new regular file is *rw-*.

 **Note:** New files and new directories continue to get their owner UID and primary group GID values set from the creating user, except when the parent directory *setgid* flag is enabled, in which case the primary group GID will be the same as the parent directory GID.

Deleting default ACLs

The process for deleting a default ACL is also the same as for deleting a standard ACL; again, preface with *d:* or use the `-d` option.

```
# setfacl -x d:u:name directory
```

This removes the default ACL that has been added in the previous example. To delete all default ACLs on a directory, use *setfacl -k /directory*. To delete all ACLs on a directory, use *setfacl -b /directory*. 

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